

Kazi Tasnim Zinat

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PhD candidate with 5+ years of AI research experience, specializing in Generative AI, Large Language Models (LLMs), and **Multimodal Models (Vision Language Models)** including at Amazon AWS. Building and evaluating models using PyTorch, TensorFlow, and Hugging Face. Proven expertise in developing novel benchmarks for VLM evaluation (EMNLP) and building multimodal applications for human-centric data analysis.

EDUCATION

University of Maryland, College Park

Aug 2019 – Present

Ph.D. Candidate in Computer Science (Advisor: Leo Zhicheng Liu)

College Park, MD

Thesis: Beyond Domain Boundaries: Enhancing Generalizability of Event Sequence Visual Analytics Research

University of Maryland, College Park

Aug 2019 – May 2024

Master of Science in Computer Science; GPA: 3.97/4

College Park, MD

Bangladesh University of Engineering & Technology

Feb 2013 – Sep 2017

Bachelor of Science in Computer Science and Engineering; GPA: 3.71/4

Dhaka, Bangladesh

EXPERIENCE

Applied Science Research Intern - Machine Learning

May 2023 - August 2023

Amazon Web Services (AWS), Bedrock Generative AI Team

Santa Clara, CA

- Engineered intervention-aware hybrid **Transformer-CNN** point-process model (event + intervention embeddings, weighted fusion, temporal encodings) trained on AWS SageMaker for long-range + local temporal patterns.
- Implemented **multi-headed self-attention** mechanisms and multi-objective training (NLL + cross-entropy + L2) with cross validation and efficient data binning; validated with experiments across simulated data and real-world logs, resulting a publication (P2).
- Achieved 88% RMSE reductions on industrial maintenance logs and $\sim 50\%$ lower negative log likelihood in simulated datasets compared to baseline; identified treatment-effect shifts aligned with clinical literature (P2).

Applied Science Research Intern - Machine Learning

May 2022 - August 2022

Amazon Web Services (AWS), Machine Learning Solutions Lab

Santa Clara, CA

- Formulated a theoretical framework extending Rubin's causal model for temporal point processes; extended average treatment effect beyond i.i.d. data and formalized causal relation shifts under interventions.
- Implemented counterfactual treatment effect estimator with intervention-aware propensity score matching.
- Proposed analysis protocol for average treatment effect estimation under interventions; performed ablation studies injecting three different intervention types, showing robustness of framework to intervention complexity.

Graduate Research Assistant - Human Computer Interaction

Spring 2021 – Present

University of Maryland (Human-Data Interaction Lab)

College Park, MD

- Created **ProcVQA**, the first benchmark for evaluating SOTA Vision-Language Models on structural effects of process visualizations (**image and text data**); curated 118 charts (2,583 ground-truth tuples, 144 expert-validated questions); evaluated 21 models (incl. GPT-4, Llama-4, Claude-3) to identify and categorize failure modes.
- Building a **RAG-based knowledge base** for event-sequence analysis; extracted structured data from 100+ research papers with added contextual chunking, multi-modal figure indexing and hybrid retrieval.
- Designing a 4-agent, human-in-the-loop framework, decomposing natural-language goals into executable code for event-sequence analytics: Planner→Intent→Technique→Code Executor, gated by user approvals

SELECTED PUBLICATIONS

P1. *ProcVQA: Benchmarking the Effects of Structural Properties in Mined Process Visualizations on Vision-Language Model Performance* (2025).

K.T. Zinat, S.M. Abrar, S. Saha, S. Duppala, S.N. Sakhamuri, Z. Liu. **EMNLP Findings**.

P2. *Uncovering Causal Relation Shifts in Event Sequences under Out-of-Domain Interventions* (2025).

K.T. Zinat, Y. Zhou, X. Lyu, Y. Wang, P. Xu. **ICANN**.

P3. *A Multi-Level Task Framework for Event Sequence Analysis* (2024).

K.T. Zinat, S.N. Sakhamuri, A.S. Chen, Z. Liu. **IEEE VIS (TVCG)**.

P4. *A Comparative Evaluation of Visual Summarization Techniques for Event Sequences* (2023).

K.T. Zinat, J. Yang, A. Gandhi, N. Mitra, Z. Liu. **EuroVis** (Computer Graphics Forum).

P5. *Comparing Native and Non-native English Speakers' Behaviors in Collaborative Writing* (2025).

Y. Chen, Y. Xiao, **K.T. Zinat**, N. Yamashita, G. Gao, Z. Liu. **ACM CHI**.

W1. *Evaluating VLMs as Accessibility Bridges for Process Visualizations* (2025).

K.T. Zinat, S.M. Abrar, S. Duppala, S.N. Sakhamuri, Z. Liu. **CVPR VizWiz Grand Challenge Workshop**.

W2. *Comparing Native and Non-native English Speakers' Behaviors* (2025).

Y. Chen, Y. Xiao, **K.T. Zinat**, N. Yamashita, G. Gao, Z. Liu. **NAACL In2Writing Workshop**.

Full list available at [Google Scholar](#).

SELECTED PROJECTS

ProcVQA: VLM Benchmark for Process Visualizations (P1, W1) | *VLM, Multi-modal AI, Hugging Face* 2024-25

- Evaluated 21 state-of-the-art VLMs (Llama-4, GPT-4.1, etc.) using GCP and Hugging Face evaluation pipelines with zero-shot protocol; exposed **19-56%** performance drop from single-hop to multi-hop reasoning.
- Analyzed structural effects and showed hallucinations rise steeply with edge density; documented instruction-following failures and failure modes (edge-value fabrication, node fusion, processing collapse)

Human-in-the-Loop Multi-Agent Event Sequence Analysis | *AutoGen, LangChain, Conversational AI* Ongoing

- Translating a theoretical **domain-specific task framework** (P3) into an executable framework using Autogen utilizing role-specific prompts; supports human feedback incorporation and sandboxed execution environment.
- Agents choose methods and parameters with full traceability, following structured workflow: user goal → planning/decomposition → framework mapping → technique selection → code execution → review/approval loop

Event Sequence Knowledge Base | *RAG, Document AI, LlamaIndex, Docling, FastAPI, React, Pydantic* Ongoing

- Implementing an ETL pipeline with multimodal extraction and grounded recommendations with inline citations.
- Designing **hybrid retrieval** (vector + BM25) and LLM-based generation to mitigate hallucination and return source-attributed recommendations via FastAPI backend powering a React chat interface for non-technical users.

Human Mobility Simulation with Deep Generative Models | *GAN, RL, Transformers, CNNs* 2022

- Evaluated a deep learning framework (MoveSim) combining Generative Adversarial Networks (GANs: generator using self-attention mechanisms and CNN-based discriminator designed to capture complex spatio-temporal patterns) and Reinforcement Learning (REINFORCE algorithm) to generate synthetic human mobility data.
- Assessed model performance on sparse (Foursquare) and dense (GeoLife) trajectory datasets, validating that data sparsity does not always correlate with inferior performance

Dense Retrievers for Open-Domain QA | *NLP, Transformers, Hugging Face, BERT* 2021

- Analyzed a Dense Passage Retrieval (DPR) system with a BERT dual-encoder (bi-encoder) architecture for open-domain question answering on the QANTA dataset.
- Fine-tuned the model using Hugging Face Transformers with in-batch negative training strategy to manage computational costs, evaluating retrieval accuracy using top-k (42.5%) against sparse TF-IDF baselines

Deep Learning for Batch Effect Removal in scRNA-seq | *GANs, Autoencoders, Transfer Learning, TensorFlow* 2020

- Benchmarked deep learning models (**GANs, Adversarial Autoencoders**) for batch correction and implemented a novel **WA-MAGAN** (Wasserstein loss) variant to address mode collapse and improve training stability.
- Analyzed domain-invariant transfer learning techniques, including gradient reversal layers (scDGN).

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, Java, R

AI/ML Frameworks: PyTorch, Hugging Face, TensorFlow, scikit-learn, LlamaIndex, AutoGen

Deep Learning Architectures: NLP, Generative AI, Transformers, CNNs, Multimodal Models (VLMs)

ML Concepts: Reinforcement Learning, Transfer Learning, Causal Inference, Predictive Maintenance

Web Development: React, FastAPI, HTML5/CSS3, Node.js

Data Visualization: D3.js, Tableau, Plotly, Leaflet.js

Tools & Platforms: Linux, Git, Docker, AWS SageMaker, Google Cloud

GRADUATE COURSEWORK

Multimodal Foundation Models, Foundations of Deep Learning, Causal Inference and Evaluation Methods,

Computational Linguistics, Information Visualization, Interactive Data Analytics, Advanced Numerical Optimization

SELECTED ACHIEVEMENTS & SERVICE

Conference Reviewer: IEEE VIS, EuroVis, ACM CHI, PacificVis; Special Recognition for Outstanding Reviews×3

Grants & Fellowships: AWS Cloud Credit for Research (\$4000, 2023), Jacob K. Goldhaber Travel Grant (2023),

Grace Hopper Celebration (GHC) Student Scholar (2023), Dean's Fellowship (UMD, 2019–2020)

Teaching: Served as Teaching Assistant for 7 courses, including Information Visualization, Programming Languages, Bioinformatics Algorithms, and Web Application Development with JavaScript

Mentoring: Guided 3 undergraduate and 3 masters students to conduct literature review, write research papers and develop interactive visualization applications, resulting in publications at top-tiered conferences